

ENTRIES 067-070 Balancing Sherbrooke: Growth, Retrain, Re-Score — September 4, 2026 — Status: COMPLETE

Balancing the Dataset Across All Three Chips

Sherbrooke Was the Thinnest Chip -- This Closes That Gap

Motivation

After Entries 060-066, chip dataset sizes were uneven: Kyiv ~1,510 bell circuits, Brisbane ~1,621 (after its own dedicated growth push), but Sherbrooke only 723 -- the thinnest of the three despite being one of the two original chips. Entry 067 grows Sherbrooke specifically, reusing the same methodology as Brisbane's growth (Entry 061): Aer/MPS ground truth, six hop-distance bins, atomic checkpointed writes, shuffled task order to avoid starving long-hop bins. 804 new circuits were added (723 -> 1,527 bell, plus the existing 24 chain circuits), bringing Sherbrooke roughly to parity with the other two chips.

Entry 068 — Fold and Retrain

The 804 new circuits were folded into the graph dataset (4,886 -> 5,690 graphs, still well within the 90/130 node/edge capacity -- max 67 nodes, 106 edges). Same-chip 80/20 baseline was retrained: MAE=1.07, R²=0.965, essentially unchanged from Entry 063's 1.17/0.973 at the smaller scale, confirming the added Sherbrooke data didn't destabilize the well-established same-chip fit.

Leave-One-Chip-Out Cross-Chip Results — Before / After Sherbrooke Growth

Direction	R² (Entry 063)	R² (Entry 068)	Change
kyiv+sherbrooke → brisbane	0.275	0.407	+0.132 (better)
brisbane → kyiv+sherbrooke	0.811	0.733	-0.078 (worse)

The two directions moved in opposite ways, and both are explainable rather than contradictory. Training on a richer, more balanced kyiv+sherbrooke pool gave the model more diverse noise structure to learn from, which transferred better to the untouched Brisbane test set -- a real generalization gain. The reverse direction got harder for a structural reason, not a regression: the kyiv+sherbrooke test set itself grew by 804 new, harder examples (many long-hop, previously-unsampled pairs) that a Brisbane-only-trained model had no chance to prepare for. This is the same pattern seen in Entries 061-064's non-monotonic Brisbane R² trajectory: a harder, more representative test set can look like a worse score even when nothing about the model's underlying competence changed.

Entry 069 — Combined Three-Chip Generalist, Retrained

Chip	MAE (Entry 065)	MAE (Entry 069)	floor-collapse MAE (069)
Kyiv	0.96	0.77	0.56 (n=26)
Sherbrooke	1.74	1.45	0.56 (n=40)
Brisbane	1.05	0.99	0.32 (n=4)
Overall	1.17	1.07	—

Every chip's held-out MAE improved, and Sherbrooke's floor-collapse MAE -- the hardest failure mode this project has chased since Entry 036 -- fell from 3.13 to 0.56, a direct benefit of Sherbrooke finally having enough training volume to let the floor-collapse up-weighting (Entry 051) do its job properly. The deployable weights were saved (entry069_combined_params.json) and used to re-score all 24,003 precomputed qubit pairs (Entry 070), replacing the live demo's Brisbane predictions and refreshing ground truth for the 804 newly-simulated Sherbrooke pairs.

Entries 067-070 — Conclusion

Balancing the thinnest chip's dataset produced a strictly better combined generalist model across all three chips, and a real (if partial) improvement in the hardest cross-chip generalization direction. It also produced a harder, more honest test in the other direction -- exactly the kind of result this project has learned to report as informative rather than alarming. The live demo now reflects the retrained combined model and the wider ground-truth coverage for Sherbrooke.

Research Log — Index Addendum (Entries 067-070)

Files: entry067_grow_sherbrooke.py, entry068_build_graphs.py, entry068_train_gnn.py, entry068_cross_chip.py, entry069_combined_train.py, entry070_score_all_pairs.py. Data: quantumbridge_data/entry067_sherbrooke_bell_dataset.json (804 new circuits), entry068_graph_dataset.json (5,690 graphs), entry069_combined_params.json (deployable weights), entry070_combined_lookup.json (rescored demo lookup).